

DATE:

EXPT NO:01

EARTHQUAKE ANALYSIS SYSTEM USING MACHINE LEARNING

AIM

To analyze earthquake data using Exploratory Data Analysis (EDA), visualize important earthquake patterns, preprocess the dataset, build a Machine Learning model to predict tsunami occurrence, evaluate the model, and export the processed data to Excel.

PROCEDURE

1. Download the Earthquake dataset.
2. Import the required Python libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn.
3. Load the earthquake dataset using Pandas.
4. Display the first few records, dataset shape, information, and missing values.
5. Remove unnecessary columns such as title and location columns.
6. Check and handle missing or duplicate values.
7. Analyze important features such as magnitude, depth, latitude, longitude, cdi, mmi, sig, gap, and nst.
8. Visualize the distribution of tsunami occurrences using a count plot.
9. Create a visualization to identify the relationship between magnitude and tsunami occurrence.
10. Separate the input features (X) and target variable (y), where tsunami is the target.
11. Split the dataset into training and testing data.
12. Train a Machine Learning model using the training data.
13. Predict tsunami occurrence using the test data.
14. Evaluate the model using suitable performance metrics such as Accuracy Score and Confusion Matrix.
15. Export the processed earthquake data to an Excel file.
16. Display the analysis results.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import accuracy_score, confusion_matrix

# 1. Load dataset
df = pd.read_csv("earthquake_data.csv")

# 2. Display data
print(df.head())

# 3. Basic information
print(df.shape)
print(df.info())

# 4. Check missing values
print(df.isnull().sum())

# 5. Fill missing values
df["cdi"] = df["cdi"].fillna(df["cdi"].median())
df["mmi"] = df["mmi"].fillna(df["mmi"].median())

# 6. Remove unnecessary columns
df = df.drop(["title", "location"], axis=1)

# 7. Visualization - Tsunami
sns.countplot(x="tsunami", data=df)
plt.title("Tsunami Occurrence")
plt.show()

# 8. Visualization - Magnitude vs Tsunami
sns.countplot(x="magnitude", hue="tsunami", data=df)
plt.title("Magnitude vs Tsunami")
plt.xticks(rotation=20)
plt.show()

# 9. Convert categorical data into numbers
df = pd.get_dummies(df, drop_first=True)

# 10. Separate features and target
X = df.drop("tsunami", axis=1)
y = df["tsunami"]

# 11. Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 12. Train Logistic Regression model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```

# 13. Make prediction
y_pred = model.predict(X_test)

# 14. Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# 15. Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt="d")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# 16. Export DataFrame to Excel
df.to_excel("earthquake_analysis.xlsx", index=False)

```

OUTPUT:

```

mmi alert tsunami sig net nst      title magnitude      date_time cdi
longitude      location continent      country
0  M 7.0 - 18 km SW of Malango, Solomon Islands      7.0 22-11-2022 02:03      8
7  green      1 768 us 117 0.509 17.0      mww 14.000 -9.7963 159.596
Malango, Solomon Islands Oceania Solomon Islands
1  M 6.9 - 204 km SW of Bengkulu, Indonesia      6.9 18-11-2022 13:37      4
4  green      0 735 us 99 2.229 34.0      mww 25.000 -4.9559 100.738
Bengkulu, Indonesia      NaN      NaN
2  M 7.0 -      7.0 12-11-2022 07:09      3
3  green      1 755 us 147 3.125 18.0      mww 579.000 -20.0508 -178.346
NaN Oceania      Fiji
3  M 7.3 - 205 km ESE of Neiafu, Tonga      7.3 11-11-2022 10:48      5
5  green      1 833 us 149 1.865 21.0      mww 37.000 -19.2918 -172.129
Neiafu, Tonga      NaN      NaN
4  M 6.6 -      6.6 09-11-2022 10:14      0
2  green      1 670 us 131 4.998 27.0      mww 624.464 -25.5948 178.278
NaN      NaN      NaN

```

(782, 19)

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 782 entries, 0 to 781
```

```
Data columns (total 19 columns):
```

```

#      Column      Non-Null Count  Dtype
----  -
0  title      782 non-null      object
1  magnitude  782 non-null      float64

```

2	date_time	782	non-null	object
3	cdi	782	non-null	int64
4	mmi	782	non-null	int64
5	alert	415	non-null	object
6	tsunami	782	non-null	int64
7	sig	782	non-null	int64
8	net	782	non-null	object
9	nst	782	non-null	int64
10	dmin	782	non-null	float64
11	gap	782	non-null	float64
12	magType	782	non-null	object
13	depth	782	non-null	float64
14	latitude	782	non-null	float64
15	longitude	782	non-null	float64
16	location	777	non-null	object
17	continent	206	non-null	object
18	country	484	non-null	object

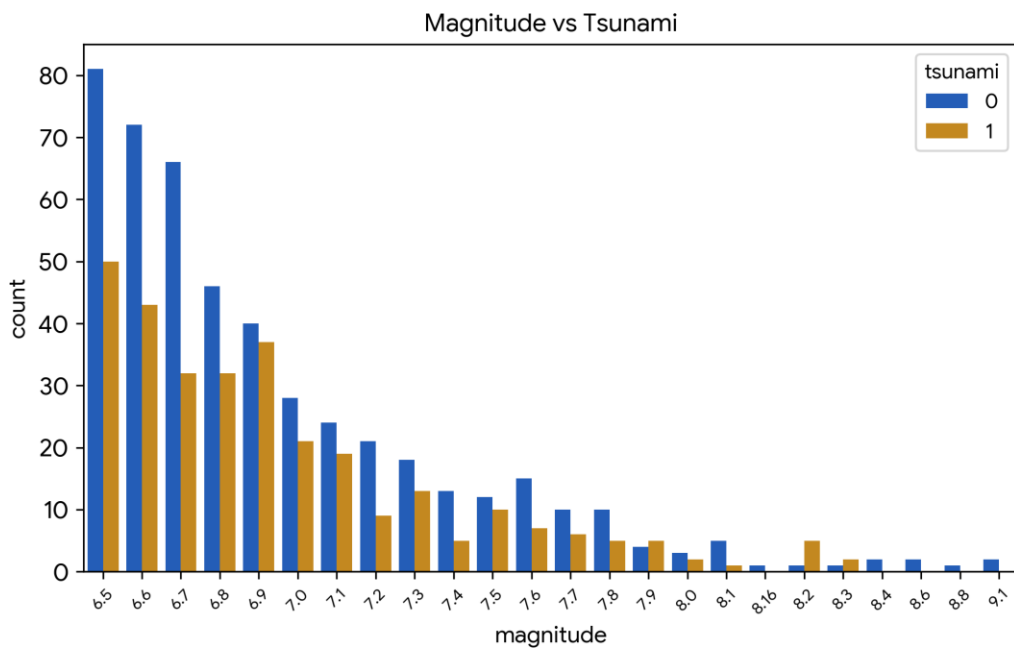
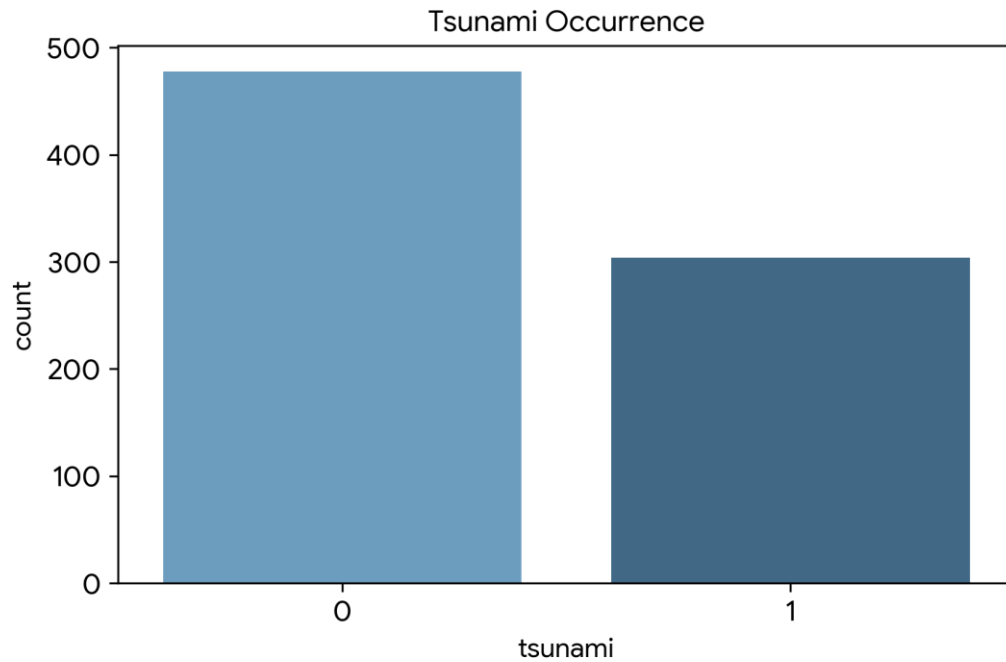
dtypes: float64(6), int64(5), object(8)

memory usage: 116.2+ KB

None

title	0
magnitude	0
date_time	0
cdi	0
mmi	0
alert	367
tsunami	0
sig	0
net	0
nst	0
dmin	0
gap	0
magType	0
depth	0
latitude	0
longitude	0
location	5
continent	576
country	298

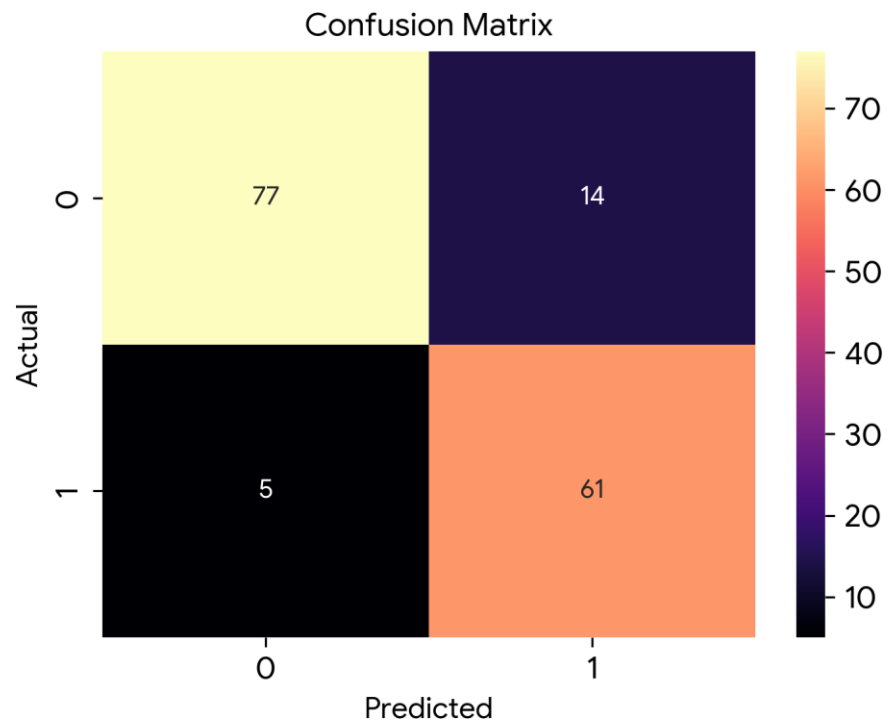
dtype: int64



```

/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_logistic.py:465:
ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.
Accuracy: 0.8726114649681529

```



RESULT: